

Mahi Aminu Aliyu

Abuja, Nigeria

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RESEARCH INTERESTS

My research focuses on causal representation learning and inductive biases – specifically, how incorporating causal structure into learned representations can improve robustness and out-of-distribution generalization, with applications in low-resource NLP and public health.

EDUCATION

M.Sc., Computer Science, Abubakar Tafawa Balewa University, Nigeria 2023 – 2026 (Expected)
Advisors: (main) Prof A. Y Gital & Dr B.I Ya'u 🌐

B.Sc., Computer Science, Al-Qalam University Katsina, Nigeria 2018 – 2023
Thesis: Water Quality Prediction Model
Advisor: Dr Mansir Abubakar 🌐

EXPERIENCE

Analysit/Programmer April 2026 – Present

Child Health and Mortality Prevention Surveillance - ATBU Bauchi

- Prepare, clean, and structure longitudinal HDSS datasets to support AI/ML research, including predictive modelling, pattern detection, and demographic analysis.
- Maintain, enhance, and support the HDSS primary data system, ensuring high availability and data accuracy.
- Collaborate with researchers, data scientists, and stakeholders to translate demographic indicators into technical requirements and reliable code.

Artificial Intelligence and Social Informatics Associate Oct 2025 – Mar 2026

Africa Hub for Innovation and Development

- Refactored WellBridge (a Tuberculosis chatbot) into class-based architecture with GPU-accelerated STT, TTS, and LLM components, improving maintainability, scalability, and reducing latency by 30%.
- Expanded its reach through WhatsApp integration and established a structured feedback loop with end-users to guide system improvements.
- Scraped and analyzed 2,000+ public health-related social media posts across four platforms to build a dataset for sentiment analysis on vaccine misinformation.

Research Collaborator 2024 – Present

ML Collective (Remote)

- Led the ML Theory and Optimization group (5 members), conducting literature-driven research on generalization techniques with a focus on causal representation learning, including experimental design, execution, and research reporting.
- Contributed to multimodal research by curating, cleaning, and organizing datasets to support reliable experimentation and analysis.
- Presented research ideas and findings at conferences and workshops, engaging with the research community and integrating feedback into ongoing work.
- Facilitated weekly Saturday reading sessions, guiding technical paper discussions and fostering critical analysis of research papers across the community.

Data Engineer & Analyst 2023 – 2025

Horizon IT Services

- Curated a low-resource Hausa dataset by extracting and cleaning 500+ text entries from Al-Barka radio station's news archives
- Applied advanced data cleaning techniques to transform 2,000+ raw Hausa news entries into a structured CSV dataset
- Collaborated on dataset enhancement by introducing categorization and optional labeling, enabling downstream use in Kaggle competitions and applications in NER, sentiment analysis, and machine translation for Hausa language

PUBLICATIONS & MANUSCRIPTS

[3] **Mahi Aminu Aliyu**, Chisom Chibuike, Fatimo Adebajo, Omokolade Awosanya, Samuel Oyeneye. When Distributions Shift: Causal Generalization for Low-Resource Languages. arXiv preprint 

[2] Mardiyah Oduwale, Prince Mireku, Fatimo Adebajo, Oluwatosin Olajide, **Mahi Aminu Aliyu**, Jekaterina Novikova. AfriCaption: Establishing a New Paradigm for Image Captioning in African Languages. In **7th Workshop on African Natural Language Processing (AfricaNLP)**, Jan 2026 

[1] **Mahi Aliyu**, Mansir Abubakar, Armaya'u Z. Umar, Alwatben Batoul Rashed, and Rukayya Musa Ismail. Analysis of machine learning models for efficient water quality prediction. In Journal of Information and Technology Management (JISTM), 10(40), Sep. 2025. 

CONFERENCES, SUMMER SCHOOLS & WORKSHOPS

Foundations of Causal Inference Workshop, Isaac Newton Institute, Cambridge – Virtual Participant Selected to attend workshop on causal inference foundations led by top global researchers from EPFL, Cambridge, Harvard, and ETH Zürich.	2026
Machine Learning Summer School - Senegal (MLSS) – Participant Selected to attend advanced ML training covering NLP, optimization, reinforcement learning, optimal transport, diffusion models, federated learning, and foundation models, led by researchers from Google, NYU, Apple, and KAUST.	2025
Deep Learning Indaba 2025, Rwanda – Attendee Selected to attend Africa’s largest AI conference with travel grant acquired from ML Collective.	2025

SELECTED PROJECTS

Cross-Lingual Aspect-Based Sentiment Analysis in African Languages - Translated SemEval-2014 to 17 African languages (afri-SemEval-2014) with aspect alignment heuristics. - Modified and fine-tuned pretrained transformer architectures (e.g., XLM-R, AfroXLM-R) for causal reasoning for aspect-based sentiment analysis tasks. - Causal models showed superior out-of-distribution (OOD) performance with fewer training steps.	
SageBot - Built AI tool that summarizes technical docs for developers, cutting review time by 15%. - Leveraged LLMs (ChatGPT, LLaMA-3), Hugging Face, and Langchain for high summarization accuracy. - Improved developer workflow efficiency by enabling 25% faster comprehension of complex documents.	
Dono Health - Led development of inclusive insurance MVP with Health Center, Vendor, and Insured modules. - Designed serverless AWS backend with Lambda, API Gateway, and CDK. - Integrated Flutterwave for payments and Africa’s Talking USSD API for voucher-based funding.	

TALKS & PRESENTATIONS

MLSS-Senegal '25 Poster presentation: "Towards Robust Generalization in African AI: Causal Inference and Domain Shift Mitigation"	2025
Deep Learning Indaba '25 Poster presentation: "Towards Robust Generalization in African AI: Causal Inference and Domain Shift Mitigation"	2025
ML Collective #25 Research Jam Oral Presentation: "Pretrained Models for Water-based Applications"	2025

COMMUNITY SERVICE

City Lead , Data Science Nigeria - Organize AI workshops, hands-on sessions, and reading groups for 150+ community members - Promote AI adoption and capacity building across underserved sectors	2024 – Present
DeepLearning.ai Ambassador - Facilitated knowledge sharing and engaged 100+ peers as an active community ambassador - Hosted "Growing an AI Community" event in Bauchi with 50+ participants - Organized and participated in 5 cross-community events to strengthen the regional AI ecosystem	2023 – Present

Lead, Google Developer Student Clubs

- Led the Google Developer Student Club (GDSC) at Al-Qalam University, Katsina. Organized and facilitated 10+ technical workshops and hands-on sessions with core-team members, reaching and training over 350 students in data science, web development, and Android development.
- Applied Google technologies to address UN SDGs, delivering solutions to 5 local communities

TECHNICAL SKILLS

Programming Languages: Python, C++, R

ML/AI Frameworks: PyTorch, TensorFlow, Hugging Face, Scikit-learn

Causal Inference & Experiment Tools: DoWhy, EconML, Weights & Biases

Web Frameworks & Tools: Django, Flask, Git/GitHub

Cloud Platforms: Microsoft Azure, AWS & GCP

REFERENCES

Prof. Abdulsalam Ya'u Gital

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Prof. MD. Muhammad Faruk Bashir

Principal Investigator, Child Health And Mortality Prevention Surveillance (CHAMPS) ATBU

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Karim Farid

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Riccardo Cadei

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